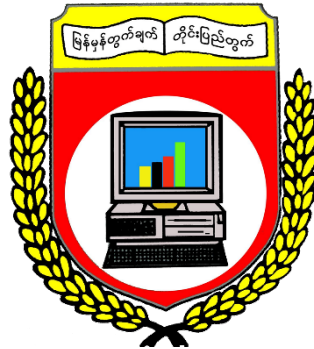


UNIVERSITY OF COMPUTER STUDIES, YANGON
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Project Report On
**AGRISENTINEL: AN EXPLAINABLE EARLY-WARNING SYSTEM
FOR NATIONAL FOOD-SECURITY RISK**

Semester IX

DATA AND KNOWLEDGE MINING (IS-212) PROJECT

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ACKNOWLEDGEMENT

Foreword I would like to thank my advisor, Dr. Hsu Myat Mo, for giving me the opportunity to work with her on this project for her support and guidance to overcome innumerable questions while doing research, and for her invaluable feedback on the way of doing research which has had a crucial influence on this project. Her quality of stating that “every modelling decision should be justified by evidence rather than assumptions” inspired the experiments in Section 3.2.4 and enriched the presentation of evaluation analysis in Chapter 4.

The motivation for my case study selection came down to the following question: Using statistical data, can we glimpse forecasting incipient food security problems? In conclusion, I want to say thanks to the support and encouragement..

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CHAPTER (1)

INTRODUCTION

Food insecurity means when people don't have regular access to adequate, safe and nutritious foods. The Food and Agriculture Organization of the United Nations (FAO) and the World Bank have released detailed statistics of the current situation of food security in different countries. These figures provide an excellent snapshot of the situation but are mostly about what is already happening. To act in time and prevent a food crisis from escalating, a country or an aid organization must be able to make forward-looking forecasts, not simply retrospective measurements.

This project "AgriSentinel" uses data mining techniques to analyze publicly available data from FAOSTAT and World Bank to forecast a country's food security risk for the upcoming year and the next five years. The project seeks to not only forecast these, but also to make the logic behind each prediction clear.

1.1 Project Background

The Food and Agriculture Organization (FAO) and the World Bank have published indicators, each in a different format and as separate datasets. FAOSTAT has a large dataset, with rows representing measurements across different countries and different periods of time, whereas the World Bank publishes different economic indicators in different datasets, with years usually in columns. None of these two data sources has all of the food security and economic indicators, and both the data sets need to be pre-processed and integrated prior to use in machine learning models.

The second is credibility. When a model is generating risk labels without explanations, users do not have enough information on which to rely for making decisions or understanding the model's predictions. AgriSentinel considers explainability as a must-have and not a nice-to-have. The project also publishes model performance scores and an explanation of the limitations of the models to give a more comprehensive and transparent assessment.

1.2 Project Objectives

The focus of this project is on four research questions:

- RQ1: What are the most nation-specific socioeconomic indicators related to hunger?
- RQ2: Can the risk level be forecasted for the upcoming year, given the conditions in the current year?
- RQ3: What are the propagation pathways of global shocks (e.g. COVID-19, 2022 price shock) in different indicators?
- RQ4: What is the amount of predictive power lost when the time horizon of the forecast is increased from 1 year to 5 years?

Based on these research questions, the project's objectives are as follows:

- The country is evaluated on three dimensions – risk level, severity of hunger, and trends – and its food security risks are predicted for one- and five-year forecast periods.
- Evaluate multiple data mining models against known benchmarks using appropriate evaluation metrics for imbalanced classes.
- Empirically validate all modeling decisions with good reason.
- Use SHAP scores and association rule mining to interpret model outputs, while transparently and honestly reporting the limitations of the results.

1.3 Project Scope and Overview

It is carried out on a national level and will involve 180 countries over 14 years (2010 to 2023) with 2,426 country years covered by eight indicators. Food insecurity at any sub-national, regional, or household level, as well as any use of satellite data, price data, or conflict data are out of the scope of this project. Our work has been organized into four different workstreams:

- Data Preparation: we create a single country-year panel from two open data sources and deal with missing data values.
- Descriptive Mining: We compute the association with hunger for each indicator, whether or not such association is statistically significant if we compare across groups, and find association rules between the indicator Hunger and the other 8 indicators.

- Predictive Mining: we train and compare classification and regression models on three different problems on two time dimensions.
- Implementation and Deployment: We make sure that our data processing workflow can be performed identically in notebooks as well as in scripts and we deploy the results as an interactive dashboard on the web.

1.4 Methodological Overview

This project follows the KDD (Knowledge Discovery in Databases) framework outlined here: Data Screening phase is when the labeled data from the FAOSTAT snapshot and World Bank snapshot sources was acquired. The Data Preparation phase was where problems like country codes were standardized and data sources combined, and missing data was dealt with. Data Transformation is when the engineered features and the classification and regression target variables are created by extrapolating forward from the trend of data at the national level for each country over two time periods.

Some of the descriptive methods (Apriori association rules, correlation measures and statistical tests of significance) and classification and regression methods are included in the Data Mining phase.

The Evaluation and Interpretation phase focuses on the model interpretability with the help of using SHAP values and secondary interpretability assessment method and discusses how the thresholds can be adjusted based on the different classification costs and risks associated with the model's predictions using a bootstrapping approach. The overall data processing pipeline is summarized in figure 2.1.

1.5 Project Benefits

The project has a number of practical applications such as the following, which are elaborated in the following sections of the book:

- Forecasts for each of the 162 countries based on two time horizons (2024, 2028) and not an analysis of current but future conditions (see Section 2.2.3).
- An explanation for each prediction in terms of SHAP values supported by a validation measure in form of a second importance estimate (see Section 4.7)

- A flexible decision threshold that enables cost control over the costs associated with missing high-risk countries vs false-positive cases (see Section 4.4)
- A public on-line interactive dashboard allowing to visualize predictions without needing any programming at all
- Reproducible results due to a fixed snapshot of the data sources, as well as a fully automatic test case that verifies that the report data can indeed be reproduced.

1.6 Chapter Summary

This chapter set out the problem the project addresses: published food-security statistics describe conditions that have already developed, so a forward-looking and explainable estimate is needed instead. Four research questions were defined, covering which indicators relate to hunger, whether next year's risk can be predicted, how global shocks propagate, and how much predictive power is lost over a longer horizon. The scope was fixed at the national level for 180 countries between 2010 and 2023, and the work was organised around the KDD process, whose stages structure the chapters that follow.

CHAPTER (2)

DATA PREPARATION AND PREPROCESSING

Data processing dominates the project. The two source datasets were intended for human readers, not for modeling; they required much reorganization before the data mining could actually begin.

This chapter is structured according to the order of the actual work done. Section 2.1 explains the two sources of data, Section 2.2 the cleaning of it, Section 2.3 the statistics of it and, finally, Section 2.4 the visual checks performed on the data. The fact that the statistics are shown with the panel, in Section 2.3, after the preparation is significant, since all the referenced numbers are derived from the cleaned panel and not from the downloaded raw data.

2.1 Data Sources

Four food security indicators were taken from the FAOSTAT food security database, and four economic indicators were taken from the world bank open data platform [1], [2]. The class files published on the world bank website group each country into a region and income group. The data snapshots used in this project are fixed to particular dates and are checksummed, so all the results shown in this book can be matched given the same inputs.

Table 2.1 summarises the project at a glance, and Figure 2.1 shows the sequence of stages the data passes through before any model is trained.

Item	Detail
Problem addressed	Estimating national food-security risk before it becomes visible
Data sources	FAOSTAT Food Security database; World Bank Open Data
Coverage	180 countries, 2010-2023 (2,426 country-year records)
Indicators	8 socio-economic indicators
Prediction tasks	Risk tier; hunger level; risk direction - at 1 and 5 years

Item	Detail
Models compared	Persistence baseline, Logistic Regression, Decision Tree, Random Forest
Explainability	SHAP values, permutation importance, Apriori association rules

Table 2. 1: AgriSentinel at a glance

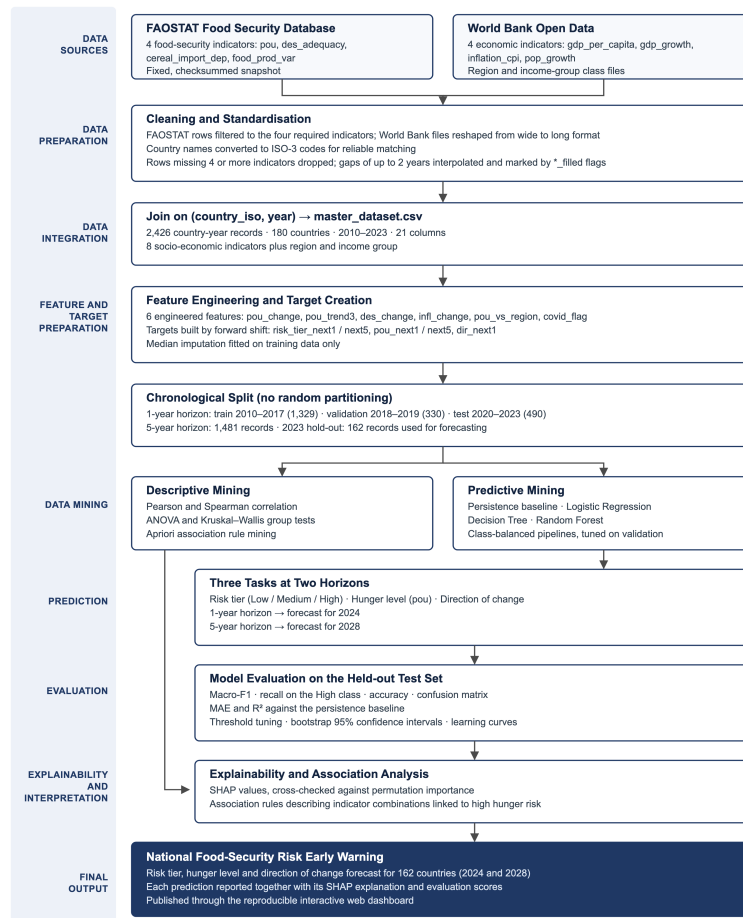


Figure 2. 1: AgriSentinel data-mining pipeline

2.1.1 Data Dictionary

Table 2.2 lists every attribute in the merged panel with its type, meaning and unit. The eight continuous indicators are the ones used as model inputs; country, year, region and income group are identifiers and labels used for grouping and reporting.

No.	Attribute	Type	Description	Unit
1	pou	Continuous	Prevalence of undernourishment	%
2	des_adequacy	Continuous	Average dietary energy supply adequacy	%
3	cereal_import_dep	Continuous	Cereal import dependency ratio	%
4	food_prod_var	Continuous	Per-capita food production variability	index
5	gdp_per_capita	Continuous	Gross domestic product per capita	USD
6	gdp_growth	Continuous	Annual GDP growth	%
7	inflation_cpi	Continuous	Inflation, consumer price index	%
8	pop_growth	Continuous	Annual population growth	%
9	country_iso, year	Categorical	Country (ISO-3 code) and year	-
10	region, income_group	Categorical	World Bank region and income classification	-

Table 2. 2: Data dictionary

2.2 Data Preprocessing

Preprocessing was carried out in four steps, each described below: missing values were handled first, the two sources were then transformed into a common shape and merged, engineered features and prediction targets were derived from the merged panel, and the result was finally split by time for model training.

2.2.1 Handling Missing Values

Missing Values Handling Steps: The missing value handling steps are as follows: During data cleaning, a row with country and year information showing 4 or more of the original 8 indicators as missing was simply killed in the data if it was too sparse (with 4 or more missing values). For short gaps (a gap = sequential years within a country's time series in which the values are missing (wikipedia 2012), but where the total number of surrounding missing years for the series (as long as it was not longer than 2 years (longer gaps require longer-term interpolation, which is akin to data manipulation, and therefore, inappropriate)), the values were interpolated between the two nearest surrounding values, then the missing cells filled according to the whether those values were flagged in the *_filled flag column. If the time period was missing for more than 2 years, those values were left "blank."

The rest of the missing values obtained after the above stripping procedure were imputed by using the median of each indicator in the modeling stage-excluding the test data in which the median was obtained. [Note that the median, but not mean, is insensitive to outliers as can be seen in Table 2.8; hence median is used here rather than mean.] The missing value rate of each indicator before this last interpolation step is given in Figure 2.2.

Figure 2.2 shows the missing rate of each indicator after cleaning. Cereal import dependency is the least complete indicator and population growth the most complete, which is why the two are treated differently in the sections that follow.

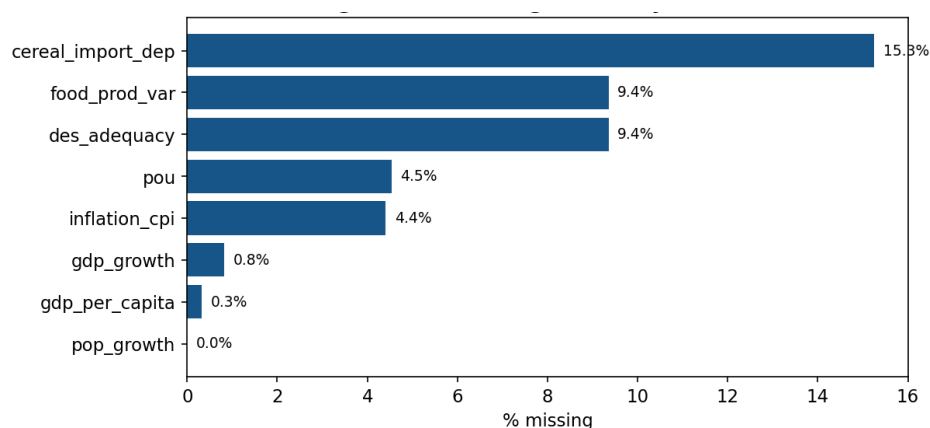


Figure 2. 2: Missing values by indicator

2.2.2 Data Transformation and Integration

Before merging these two data sources, three transformations are required. First, only rows from the FAOSTAT file that meet the four required indicators are extracted. Second, each World Bank file is converted from “wide format” (i.e., years presented in columns) to “long format” (i.e., each year on a separate row). Third, each country name is converted to a three-letter ISO code to ensure reliable matching between data sources; this step requires manual processing of names that map to multiple codes and are ambiguous, and a regression test was developed specifically for this issue. Subsequently, the three data sources were joined based on the (country_iso, year) key-value pair.

2.2.3 Feature Engineering and Target Creation

The combined data set was used to extract six features to provide the model with sufficient data, since two countries can have the same level of malnutrition in a given year, but be moving in opposite directions. The values of the extracted features for this real world data set are shown in Table 2.3.

Derived feature	Value (AFG, 2015)	Definition
pou_change	+0.6	Change in pou from the previous year
pou_trend3	+0.25	Slope of pou over the previous three years
des_change	0.0	Change in dietary adequacy from the previous year
infl_change	-5.34	Change in inflation from the previous year
pou_vs_region	+12.35	This country's pou minus its region's mean for that year
covid_flag	0	1 for the years 2020 and 2021, otherwise 0

Table 2. 3: Engineered features (Afghanistan, 2015)

Later, data series for each country were extrapolated into the future to create the predicted targets. The attributes that have been given for 2015 are not the 2015 indicators, but rather the actualisation of the 2015 target attributes for 2015 or the 2020 target attributes for 2015. This is

the case for one of the rows in table 2.4. It is checked automatically that the next level for one of the rows should always be equal to the current level in the following row, otherwise the update procedure stops and an error will be generated.

Target column	Value (AFG, row = 2015)	Source of the value
risk_tier_next1	High	Afghanistan's actual risk tier in 2016
pou_next1	20.5	Afghanistan's actual pou in 2016
dir_next1	Worsening	pou rose by more than 0.5 pp from 2015 to 2016
risk_tier_next5	High	Afghanistan's actual risk tier in 2020
pou_next5	28.9	Afghanistan's actual pou in 2020

Table 2. 4: Prediction targets (Afghanistan, 2015)

Due to the dataset ending in 2023, the records from 2023 are particularly special. All attribute fields are filled and the target fields are empty because the years that they refer to are not in the data. These 162 records are neither used in training or testing, but those records are the ones used to apply the trained model, which yields the published forecast for 2024 and 2028 on the dashboard.

Column	AFG 2023	Status
pou	29.7	Feature - present
risk_tier_current	High	Feature - present
pou_next1	(empty)	Target - 2024 not in the dataset

Column	AFG 2023	Status
risk_tier_next1	(empty)	Target - 2024 not in the dataset

Table 2. 5: A 2023 row - features complete, targets empty

2.2.4 Chronological Data Splitting

As this is time-series panel data, and a hourly-cross-section data for every year, so it is never randomly partitioned. For tasks of one-year blocks, the model is trained with data of 2010-2017, validated with 2018-2019, and tested with 2020-2022. If data of a country in the model training portion is sampled later than the time index used for the model testing, the model will be unnecessarily trained on that country's later data and evaluated on that country's earlier data, which results in a high score. Table 2.6 shows the differences in the number of lines before and after target generation and data .

File	Rows	Note
master_dataset.csv	2,426	All country-year records, before targets
features_lagged_1yr.csv	2,149	Records with a valid one-year target
train split	1,329	2010-2017
validation split	330	2018-2019
test split	490	2020-2022
features_lagged_5yr.csv	1,481	Records with a valid five-year target
features_demo_latest.csv	162	2023 only - features present, targets empty

Table 2. 6: Row counts through target creation and splitting



Figure 2. 3: Chronological train / validation / test split (1-year horizon)

2.3 Dataset Statistics

With preparation complete, the resulting panel can be described. The statistics below are all measured on the cleaned country-year panel, so they reflect the data the models actually see.

2.3.1 Dataset Size and Coverage

Table 2.7 gives the size of the finished panel: how many country-year records it holds, how many countries and years it spans, and how many regions and income groups those countries fall into.

Property	Value
Total records (rows)	2,426
Total columns	21
Countries	180
Years covered	2010 - 2023 (14 years)
Regions	7
Income groups	4

Table 2. 7: Dataset size and coverage

2.3.2 Descriptive Statistics and Outliers

Table 2.8 reports the descriptive statistics of the eight indicators. Several are strongly skewed. Inflation has a skew of 16.68, cereal import dependency has a skew of -7.77 because large crop exporters record large negative values, and GDP per capita has a skew of 2.4. This departure from a symmetric distribution is a finding that affects model choice later, and it is confirmed by the normality tests in Section 2.4.

Indicator	Mean	Std	Min	Median	Max	Skew
pou	9.75	10.05	2.5	5.7	72.5	2
des_adequacy	122.44	14.68	71	123	167	-0.16
cereal_import_dep	26.92	87.97	-1894.1	40.15	100	-7.77
food_prod_var	35.68	26.67	1	30	289	3.32
gdp_per_capita	14128.79	19927.16	219.2	5775.35	134965.82	2.4
gdp_growth	3.11	5.58	-50.34	3.32	86.83	0.9
inflation_cpi	6.43	19.17	-4.64	3.38	557.2	16.68
pop_growth	1.26	1.45	-10.93	1.2	14.23	0.24

Table 2. 8: Descriptive statistics of the indicators

Figure 2.4 shows the same eight indicators as histograms, which makes the skew visible as a shape rather than as a single number.

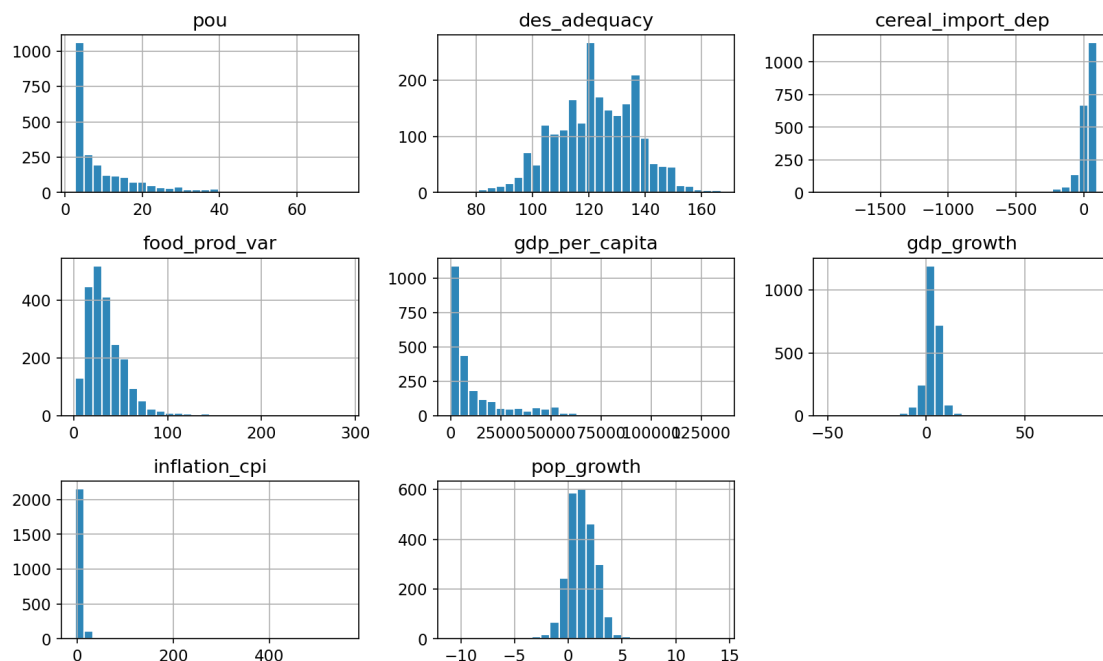


Figure 2. 4: Distribution of the eight indicators

Outliers were counted using the 1.5 times IQR rule and reported in Table 2.9, but they were deliberately retained. The extreme values are real events - the maximum recorded inflation is 557.2 percent - and these are precisely the high-risk situations the system is intended to detect. Removing them would remove the cases of interest.

Indicator	Outliers	%	Min	Max
pou	127	5.5%	2.5	72.5
des_adequacy	15	0.7%	71	167
cereal_import_dep	99	4.8%	-1894.1	100
food_prod_var	89	4%	1	289
gdp_per_capita	285	11.8%	219.2	134965.8
gdp_growth	164	6.8%	-50.3	86.8
inflation_cpi	162	7%	-4.6	557.2
pop_growth	43	1.8%	-10.9	14.2

Table 2. 9: Outliers per indicator (1.5 x IQR rule)

2.3.3 Distribution of Targets by Category

Each country-year data set was sorted into the relevant risk categories according to the malnutrition rate: “low” ($<5\%$), “medium” (between 5 and 15%) and “high” ($\geq 15\%$). The outcome of this is that categories are not equal in number: the category “high” is the smallest, yet the category that requires most distinguishing. Consequently, macro-level average indicators have been used throughout the entire evaluation process rather than merely accuracy rates.

Tier	Count	Percentage
Low	1070	46.2%
Medium	754	32.6%
High	492	21.2%

Table 2. 10: Risk-tier class distribution

2.4 Data Visualization

The figures below examine the cleaned panel from four angles: how the indicators relate to one another, how hunger differs across regions and income groups, how it has moved over time, and whether the indicators follow a normal distribution.

Figure 2.5 shows the years malnutrition path for a given set of countries. The correlation between malnutrition and dietary adequacy is the highest (Pearson correlation coefficient= -0.83): Countries that consume substantially more calories relative to their needs tend to have less malnutrition.

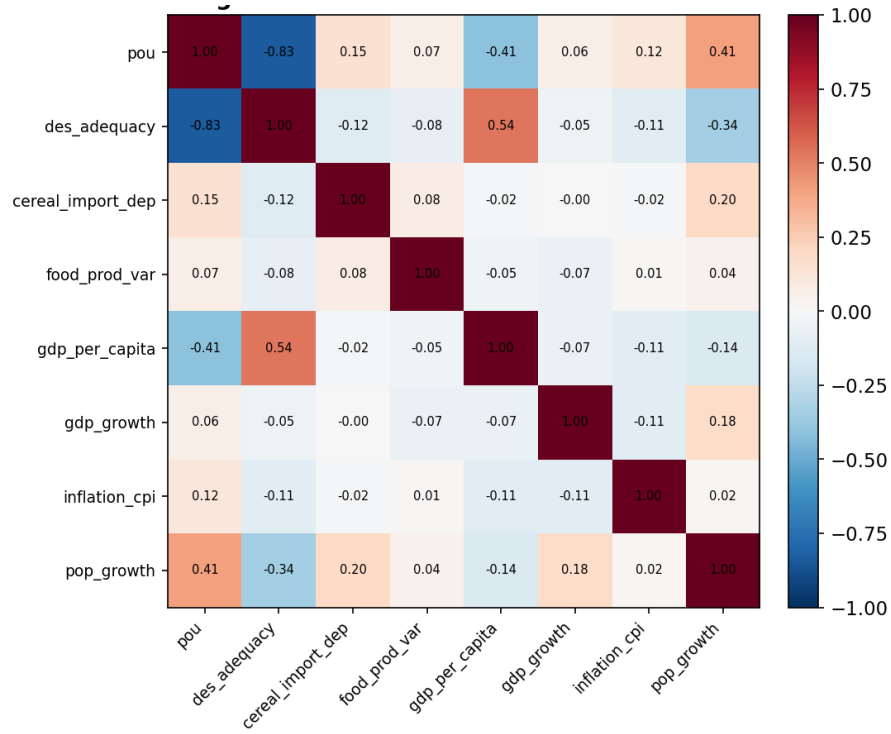


Figure 2. 5: Correlation between indicators

The variation in the levels of malnutrition among the regions in Figures 2.6 and 2.7 are statistically significant and tested in Section 3.1.2. They are not just be seen simply because of the graphs.

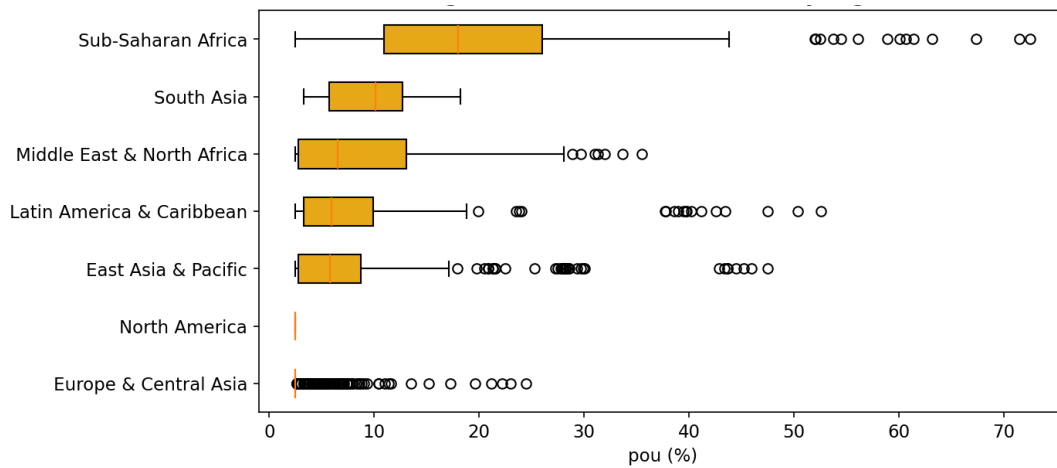


Figure 2. 6: Undernourishment by region

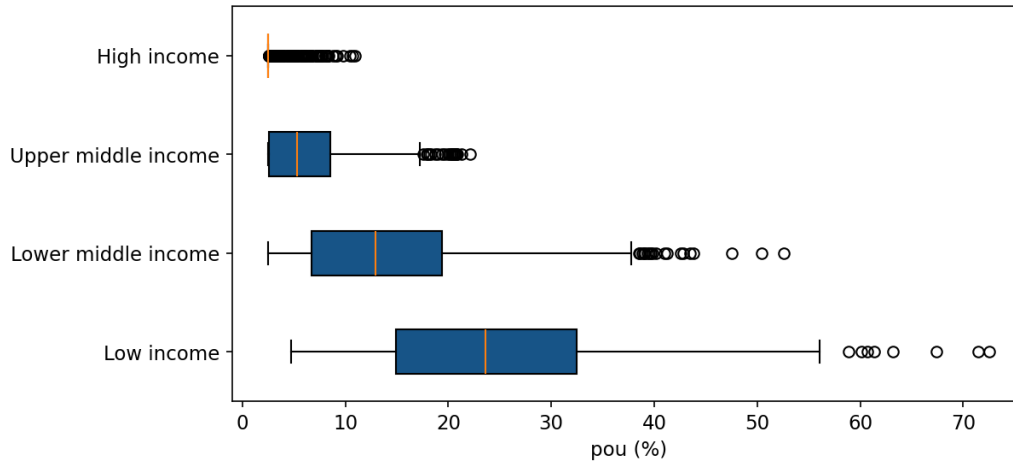


Figure 2. 7: Undernourishment by income group

Figure 2.8 illustrates the annual malnutrition trajectory for a particular set of countries. It may be hard to beat a “persistent baseline”– using the last year’s values as- represented here by the dashed grey line- within this one-year forecast window (a point that will be relevant when Chapter 4 discusses other metrics).

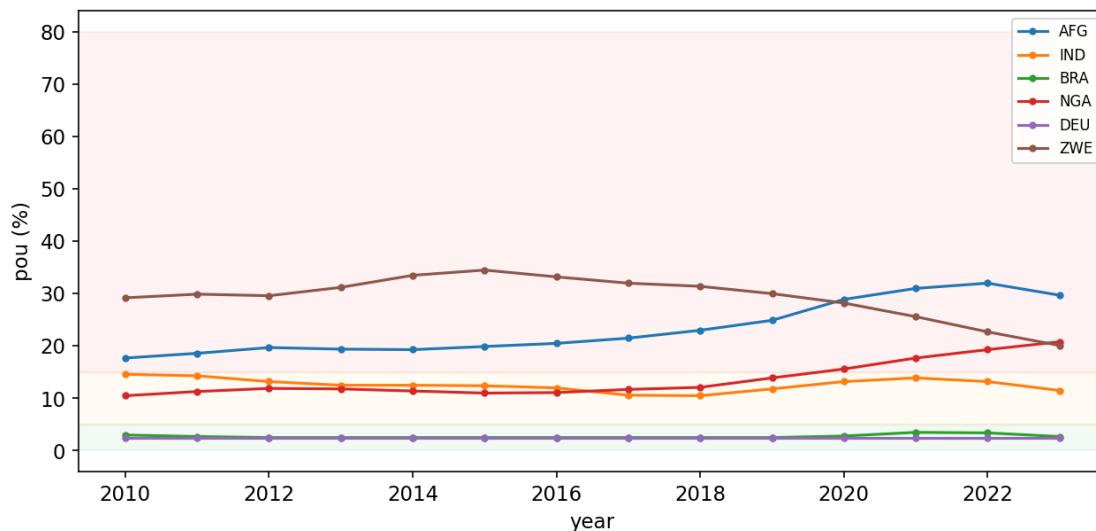


Figure 2. 8: Undernourishment trend by country (2010-2023)

Figure 2.9 confirms the shape of the distribution directly. Q-Q plots reveal obvious evidence of non-normality; a D'Agostino test leads to a rejection of normality at the 5 per cent level for all six indicators. This both justifies the use of tree-based models, which are distribution-free, and suggest the motivation for the distribution-free significance test of Section 3.1.2.

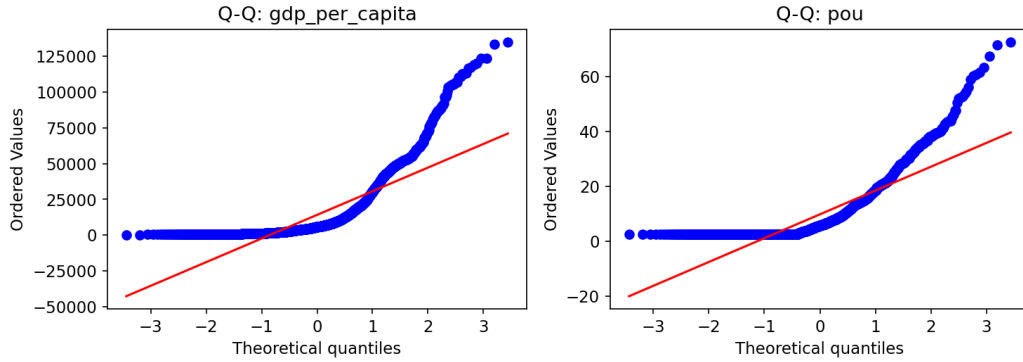


Figure 2. 9: Normality check (Q-Q plots)

2.5 Chapter Summary

This chapter prepared the data in the order the work was actually carried out. Two open sources, FAOSTAT and the World Bank, were reshaped into a common form, matched on standardised country codes and merged into a single country-year panel of 2,426 records covering 180 countries. Missing values were handled in two stages, engineered features and the three prediction targets were derived, and the panel was split by time rather than at random so that no model can be trained on years later than those it is tested on. The statistics and visual checks reported at the end of the chapter all describe this cleaned panel, and they establish two facts the later chapters rely on: the indicators are strongly skewed, and the risk classes are imbalanced.

CHAPTER (3)

APPLIED METHODOLOGIES AND MINING

The idea of "mining" will be discussed in two parts in this chapter. Descriptive mining was added for study of data structure in class preparation without the ability of prediction. In contrast, analysis of mine was used in the training model and get a model by making experimentation.

3.1 Descriptive Mining

Descriptive mining looks for structure that is already present in the data. Three questions are asked in turn: how strongly each indicator relates to hunger, whether the differences between groups of countries are statistically real, and which combinations of conditions tend to occur together.

3.1.1 Correlation Analysis: Linear and Monotonic

Using a single correlation as a measure of relationships simply reminds the reader that a relationship exists. For the assessment of relationships between variables, the Pearson correlation coefficient (the Spearman correlation coefficient, for example) can be used. In this study two coefficients were used to measure the relationship between each of the indicators of malnutrition. The relationship observed was monotonic but non-linear as the two coefficients were not equivalent.

For per capita gross domestic product, the correlation coefficients were -0.411 and -0.799 with a difference of 0.388 . These coefficients, although highly correlated, are curvilinear in nature. In this extremely poor country, as it began to escape extreme poverty through consumed consumption, the malnutrition rate decreases and eventually stopped among the rest of the wealthier nations (this can be seen when the two correlation curves are put together in a graph).

This particular relationship form was the one used that developed the tree-based model in Sec. 3.2 as this form is not strictly linear and cannot be made strictly linear using any known transformation.

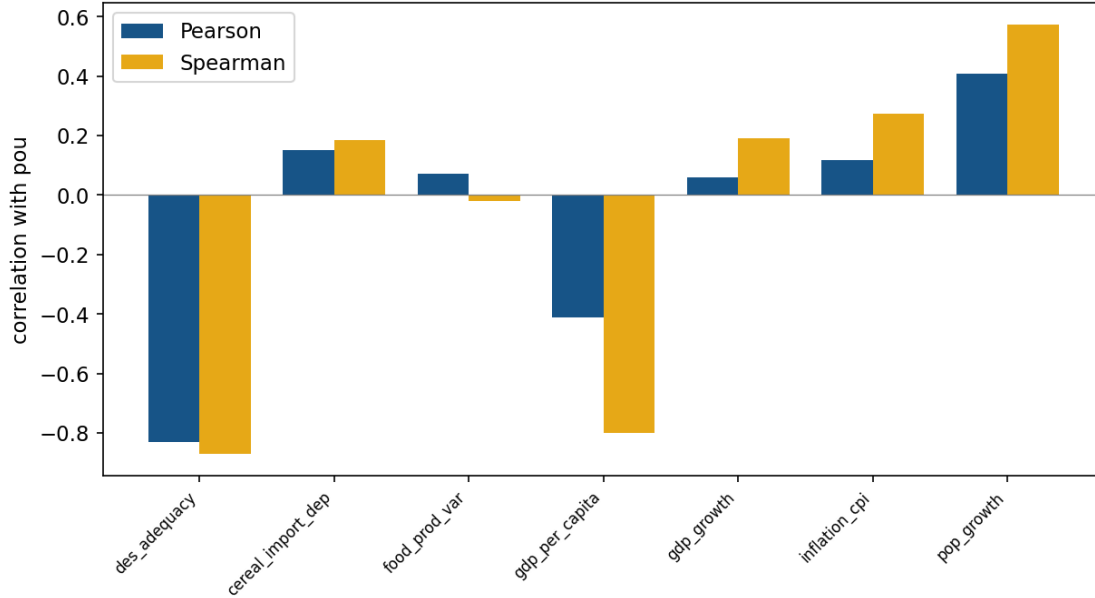


Figure 3. 1: Pearson vs Spearman correlation with hunger

3.1.2 Group Significance Testing

These As illustrated in Figures 2.6 and 2.7, it cannot be attributed solely to sampling bias that the variability across regions and income groups is not the same. In fact, we have been able to corroborate this statistically. By performing an analysis of variance (one-way ANOVA), we examined the importance of the differences in mean values of groups. For the purpose of examining the difference of medians, we carried out a similar analysis (nonparametric method requiring no assumptions regarding the distribution of the data) using the Kruskal-Wallis analysis of variance.

It was found that the p-values for the two-factor data were less than 0.05 in all the cases.

Considering the result of the normality test (Section 2.4), we have statistics that confirm that the trends in data observed above are statistically important.

Grouping	ANOVA p-value	Kruskal-Wallis p-value	Significant at 5%
Region	2.82e-237	3.69e-278	Yes
Income group	0.00e+0	0.00e+0	Yes

Table 3. 1: Group-difference significance tests

3.1.3 Association Rule Mining

The indicators were recoded as binary items e.g. HUNGER_HIGH, INCOME_LOW, SUPPLY_INADEQUATE and used the Apriori algorithm to identify combinations that co-occur. 52 rules were created. Severe hunger was the outcome in the strongest rules found are shown in Table 3.2. Two had the maximum value of 1.000 for confidence and a high lift of 5.088 indicating that in the dataset every case with the antecedent in the rule also has severe hunger, and co-occurs around 5 times more often than to be expected .

Antecedent	Consequent	Support	Confidence	Lift
INCOME_LOW + SUPPLY_INADEQUATE	HUNGER_HIGH	0.057	1.000	5.088
SUPPLY_INADEQUATE	HUNGER_HIGH	0.063	1.000	5.088
INCOME_LOW + POPGROWTH_HIGH	HUNGER_HIGH	0.120	0.613	3.119
INCOME_LOW	HUNGER_HIGH	0.160	0.511	2.598
POPGROWTH_HIGH	HUNGER_HIGH	0.135	0.498	2.533

Table 3. 2: Strongest association rules for serious hunger

Indicators that are only visible in conjunction with other indicators can be much more valuable as a message than a sole indicator. For example, the correlation between hunger and grain import dependency alone is weak (Pearson correlation = 0.151), because high income, import-dependent countries tend to have plenty of food and this indicator becomes significant only in the context of low income. If only univariate analysis is performed this interaction is completely missed.

3.2 Predictive Mining

Predictive mining moves from describing the data to forecasting from it. All models in this section are implemented with the scikit-learn library [3]. The section sets out the algorithms applied, how they were configured, the baseline they must beat, the design decisions that were settled by experiment rather than assumption, and the model finally proposed.

3.2.1 Applied Algorithms

Four different types of models were used for the task of classifying hunger levels: Persistence (predicting that next year's level will be the same as this year's) was used as a simple baseline. As hunger levels change slowly, this is not only a symbolic benchmark, but also a truly robust one. Logistic regression was used as a linear benchmark.

The rule weights of a decision tree are explicitly interpreted.

Random Forests, combining many decision trees, can model the curvilinear relation as found in section 3.1.1. We also used Random Forest regression in the hunger level prediction task and multi-indicator task.

Model	Type	Role in this project
Persistence	Baseline	The score every trained model must exceed
Logistic Regression	Classification	Linear reference model
Decision Tree	Classification	Directly interpretable rules
Random Forest	Classification and regression	Captures non-linear structure; used for all three tasks

Table 3. 3: Models applied in this project

3.2.2 Model Setup

All models are configured as pipelines; preprocessing is only carried out on training. Each pipeline start with median interpolation. Logistic regression pipeline additionally contain standardize step because linear model is very sensitive with order of magnitude - e.g there is one magnitude of difference between the per capital gdp of 1000s of dollars and that of population growth in percentage of a given country-decision tree based models would not need this step.

All of these classifiers using class balanced weights.

This decision is validated through section 3.2.4. All running parameters are stored in the configuration file instead of coded in the running code - this means that it only need 1 single line

to change parameters including random seed 42, the number of trees and the year-which decides where to split the data based on the time-when comparing time-wise models.

3.2.3 Baseline Models

Persistence baseline model achieve the score of 0.963 on a one-year-span test set and 0.823 on a five-year-span test set in the context of the marco F1 score. All of these results is extremely high if considering that the model does not learn anything during training and provide very hard baseline to reach. The use of reporting results can be considered as giving post-trained model a better-than-actual looking value.

3.2.4 Empirically Validated Design Decisions

There were four design choices driven by data not hypothesis. Class imbalance. Only 21.2% of records fall into high-risk group but are associated with highest cost of false negatives. We validated the same logistic regression model two times using default weights versus class-balanced weights, showing a change of recall from 0.953 to 0.974 with a drop of the marco F1, since warning early for high risk countries is more important than preventing early warnings from the medium-risk one so we choose class-balanced weights for all of the classifier.

Setting	Macro-F1	Recall on High class
Default class weights	0.971	0.953
Balanced class weights	0.967	0.974

Table 3. 4: Effect of handling class imbalance

Class weighting, continued. Table 3.4 answers whether balancing helps at all; it does not say how much weight the High tier should carry. To settle that, five settings were swept on the validation split - no weighting, the automatic balanced setting, and High weighted two, three and five times - and both sides of the trade were measured, because raising the weight buys recall at the cost of precision. The sweep was repeated for each model family, since they do not respond alike.

Model	class_weight	Macro-F1	Precision (High)	Recall (High)	F2 (High)
Logistic Regression	none	0.948	1.000	0.956	0.964
Logistic Regression	balanced	0.957	0.985	0.985	0.985
Logistic Regression	High x2	0.953	0.986	1.000	0.997
Logistic Regression	High x3	0.948	0.971	1.000	0.994
Logistic Regression	High x5	0.942	0.958	1.000	0.991
Random Forest	none	0.958	0.985	0.941	0.950
Random Forest	balanced	0.952	0.970	0.941	0.947
Random Forest	High x2	0.952	0.970	0.941	0.947
Random Forest	High x3	0.954	0.970	0.941	0.947
Random Forest	High x5	0.954	0.970	0.941	0.947

Table 3. 5: Effect of different class weights on the High tier (1-year horizon, validation split)

The two families behave very differently. For Logistic Regression the weighting works exactly as intended: recall on the High tier rises from 0.956 with no weighting to 1.000 once High is weighted twice, while precision only falls from 1.000 to 0.986, and the best F2 of 0.997 is reached at High x2. Weighting beyond that point buys no further recall and only costs precision. The Random Forest barely moves at all, holding recall at 0.941 whatever weight it is given, because a deep forest on a well-separated signal is already confident enough that re-weighting changes few of its decisions. That difference is itself a result: the weighting is a useful control for the linear model and an ineffective one for the forest.

Feature selection. The*filled*flag columns denote how the data was derived, not the condition of the country. It was not certain whether they should be passed to the model. An ablation was performed comparing the same Random Forests cross-validated first with the eight predictors only, then with the 8 predictors + the 8 corresponding flags. The macro-F1 remained the same in each case (0.9989), and so they were removed from the models, kept only as a trace..

Feature set	Macro-F1 (5-fold cross-validation)
Features only	0.9989
Features plus *_filled flags	0.9989 (difference 0.0)

Table 3. 6: Flag-ablation experiment

Interpretable regression. A linear least squares regression was used with undernourishment as the outcome, using the seven indicators remaining apart from the target feature. The amount of variance explained is 68.6%. Of the predictors, only dietary adequacy and population growth are significant at the 5% level, the other five indicators are not significantly contributing once all the others are controlled for. The result corroborates that from the correlation analysis and also provides a parallel interpretable complement to the trees model..

Term	Coefficient	p-value	Significant at 5%
const	74.2664	< 0.001	Yes
des_adequacy	-0.5401	< 0.001	Yes
cereal_import_dep	0.0025	0.094	No
food_prod_var	0.0007	0.875	No
gdp_per_capita	0.0000	0.175	No
gdp_growth	-0.0141	0.523	No
inflation_cpi	0.0090	0.146	No
pop_growth	1.0868	< 0.001	Yes

Table 3. 7: OLS regression coefficients (target = pou, R-squared = 0.686)

Regularisation of the Random Forest. We performed a grid-search over tree depth and minimum leaf size on the training set alone. The top of the one-year grid is presented in Table 3.8. As the trees are more restricted, we can see that the macro-F1 for training and validation does not improve, and this means the gap cannot be explained by over fitting. We take the values that have the highest macro-F1 for the validation data.

max_depth	min_samples_leaf	Train macro-F1	Validation macro-F1
None	2	0.995	0.956
15	2	0.995	0.956
8	2	0.990	0.956
10	2	0.994	0.952
None	10	0.971	0.949

Table 3. 8: Random Forest regularisation grid (1-year horizon, top rows by validation score)

3.2.5 Proposed Model

We propose the Random Forest for all three. At both horizons, it yields the highest macro-F1 for risk-tier and captures the curvilinearGDPrelation discovered in section 3.1.1 without any transformation. Further it supports explanations by the exact tree-based estimator for SHAP. It has limitations (honest reporting in Chapter 4): at 1-year horizon statistically insignificant outperformance of persistency.

3.3 Chapter Summary

This chapter applied the mining itself, in two parts. Descriptive mining measured how each indicator relates to hunger, confirmed with formal tests that the differences between regions and income groups are statistically real rather than accidental, and used Apriori to surface combinations of conditions that occur together with serious hunger. Predictive mining then trained four model families against a persistence baseline and, rather than assuming the right configuration, settled each design decision with a measured experiment covering class balancing, the weight carried by the High tier, feature selection and regularisation. The Random Forest was proposed on that evidence, with its limitations left to be reported in the evaluation.

CHAPTER (4)

EVALUATION

Evaluation proceeds from the headline scores to the evidence behind them. Sections 4.1 and 4.2 define the metrics and compare the models, Sections 4.3 to 4.6 test how far those scores can be trusted, Section 4.7 explains what drives them, and Sections 4.8 and 4.9 report the remaining tasks and the shock analysis before the findings are drawn together in Section 4.10.

4.1 Performance Metrics

Since the proportions within each class were unbalanced the accuracy measure could not be used: we would have a good value even if a model always predict category "high-risk". We can then assume that it would fail the true "high-risk" countries. So, we will consider only macro-average F1- score (the three categories weight 1/3) to evaluate all three models.

The recall of the "high-risk" category will be computed as well since this is the class having largest costs if fails.

The absolute error, the R^2 to persistence will be report for the regression task. Finally once all of the steps are done, we compute a precision score for the test set.

Macro-F1 treats a false alarm and a missed warning as equally serious, which does not match this problem: failing to flag a country that turns out to be high-risk is the more costly error. The F-beta score makes that weighting explicit. Beta equal to one reproduces macro-F1 exactly, values below one favour precision and values above one favour recall, so F2 - recall weighted twice as heavily as precision - is reported alongside macro-F1 throughout Section 4.2.

4.2 Results Comparison

Table 4.1 presents the test set performances for the four models under a period of prediction of 1 year and 5 year. The best macro F 1-score were achieved by Random Forest, both for the 1 year prediction (0.965) and the 5 year prediction (0.865). All trained models perform best on the persistence over the 5 year prediction, contrary to 1 year prediction for which difference is very small (0.965 vs 0.963). We will test in section 4.5 if these are statistically significant or not.

Model	Horizon	Accuracy	Macro-F1	Recall on High
Persistence	1yr	0.965	0.963	0.963
Logistic Regression	1yr	0.957	0.956	0.991
Decision Tree	1yr	0.953	0.947	0.954
Random Forest	1yr	0.967	0.965	0.963
Persistence	5yr	0.84	0.823	0.761
Logistic Regression	5yr	0.852	0.833	0.789
Decision Tree	5yr	0.852	0.842	0.775
Random Forest	5yr	0.877	0.865	0.803

Table 4. 1: Risk-tier model comparison on the test set

Table 4.2 repeats that comparison under the recall-weighted metric. Because beta equal to one is macro-F1, the two tables can be read side by side: the macro-F1 column is the same score as in Table 4.1, and the F2 columns show what happens once a missed warning is counted twice as heavily as a false alarm.

Model	Horizon	Macro-F1	Macro-F2	F2 (High tier)
Persistence	1yr	0.963	0.963	0.965
Logistic Regression	1yr	0.956	0.958	0.984
Decision Tree	1yr	0.947	0.948	0.949
Random Forest	1yr	0.965	0.965	0.965
Persistence	5yr	0.823	0.821	0.778
Logistic Regression	5yr	0.833	0.828	0.805
Decision Tree	5yr	0.842	0.841	0.797
Random Forest	5yr	0.865	0.861	0.826

Table 4. 2: Recall-weighted comparison (F2) on the test set

The ranking is not identical under the two metrics. Random Forest still has the highest macro-F1 at one year, but Logistic Regression reaches an F2 of 0.984 on the High tier against Random Forest's 0.965, so a system judged only on how many high-risk countries it catches would prefer the simpler model. At five years Random Forest leads on both metrics. Reporting F2 alongside macro-F1 therefore changes what can be claimed, which is the reason for including it rather than relying on a single metric.

4.3 Confusion Matrix Analysis

Figure 4.1 the one year confusion matrix of random forest. Table 4.3 list classification performances. The most important one thing is that, whichever predicted periods, no highest risk class was classified as a lowest risk class. In warning purpose of country risks, the most harmful classification would be assigning the highest risk state to the lowest risk one, leading country owners feel confident with their severe potential risk state. In this work, errors are between adjacent classes rather than among extremes in addition the harms are not large.

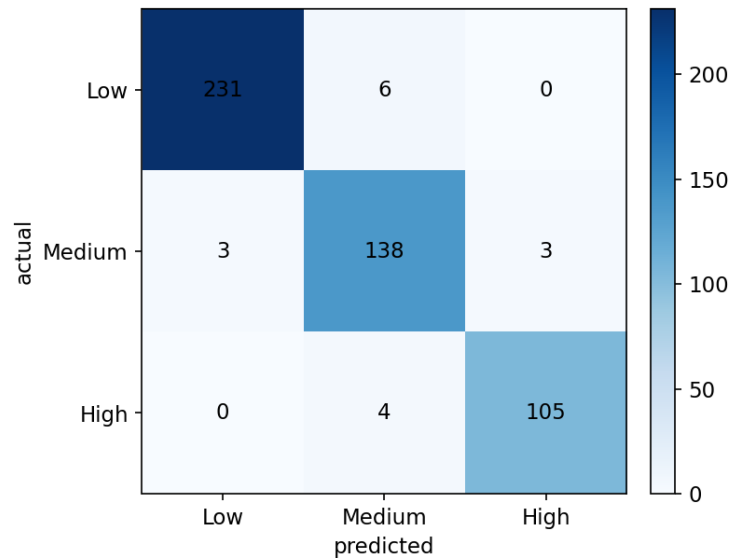


Figure 4. 1: Confusion matrix (Random Forest, 1-year risk tier)

Horizon	Class	Precision	Recall	F1-score	N
1 year	Low	0.987	0.975	0.981	237
1 year	Medium	0.932	0.958	0.945	144

Horizon	Class	Precision	Recall	F1-score	N
1 year	High	0.972	0.963	0.968	109
5 years	Low	0.931	0.931	0.931	159
5 years	Medium	0.762	0.842	0.800	95
5 years	High	0.934	0.803	0.864	71

Table 4. 3: Per-class results (Random Forest)

4.4 Threshold Tuning

The classifier defaults to assign samples into the highest probability class, this means, implicitly, every misclassification is given the same penalty. However, this is not the case in this research project. So only for "high" class, performed threshold scan upon validation set: Table 4.4 show on year threshold scan results.

Recall at "High" was 1.000 with threshold lowered to 0.25 as recall was constant and precision was 0.971. Then we chosen threshold as 0.25 as lowest threshold while precision remains 0.70. Applied this threshold on test set. We recall at "High" with 0.982, and precision with 0.947..

Threshold	Recall on High (validation)	Precision on High (validation)
0.25	1.000	0.971
0.30	1.000	0.986
0.35	1.000	0.986
0.40	1.000	1.000
0.45	1.000	1.000
0.50	0.985	1.000
0.55	0.985	1.000

Table 4. 4: Validation threshold sweep (1-year horizon)

This threshold tuning can be a practical contribution of the project, operator can choose its preferred operating point according to the value-weight assigned to "false negatives" and "false

positives". Also, impacts on choices are from this empirically experiment study, not arbitrary assumptions.

4.5 Bootstrap Confidence Intervals

Since this average alone cannot determine how much of the gap between the two models is due to specific country-year samples in the test set, samples were drawn 1000 times from the test set with replacement for both the Random Forest model and the persistence baseline and 95% confidence intervals were calculated for both.

Horizon	Random Forest 95% CI	Persistence baseline 95% CI
1 year	[0.946, 0.979]	[0.944, 0.978]
5 years	[0.823, 0.903]	[0.776, 0.863]

Table 4. 5: Bootstrap 95% confidence intervals for macro-F1

Both times dimensions have confidence intervals that overlap; at the chosen test set the score difference between the Random Forest and persistence models is not statistically significant, thus the apparent score of 0.965 must be recognized as evidence that the model does not substantially perform better than a guess derived from predicting last year's grading. While this is a clear weakness that will be mentioned here rather than be omitted, the actual value of the learned model is not reflected by this score, but is reflected in three qualities the baseline lacks: ability to produce an explanation for any single prediction (Section 4.7), flexible confidence threshold (4.4), and inability to mistake high and low for each other (4.3). Even more obvious is this over a 5-year forecast horizon with its larger time gaps and baseline model's much reduced score.

4.6 Learning Curves

Learning curves have the desirable ability to help us identify when models are hitting their capacity ceilings, or when performance is limited by training data. In Table 4.6 we have learning curve results for macro-F1 for validation set, as the model size is increased.

Horizon	Training size	Train macro-F1	Validation macro-F1
1 year	331	1.000	0.922

Horizon	Training size	Train macro-F1	Validation macro-F1
1 year	774	1.000	0.956
1 year	1,106	1.000	0.954
5 years	231	1.000	0.733
5 years	539	1.000	0.790
5 years	770	1.000	0.817

Table 4. 6: Learning-curve summary

On the single year prediction horizon, the validation set score appears to have plateaued, meaning more records would be unlikely to boost the score, and on the 5-year prediction horizon, the score continues to grow at maximum training data available. This indicates the 5-year model faces limitations from lack of data (having only 14 available historical country data years means the training data is far sparser for 5-year samples than other periods), this is a concrete quantitative justification compared to the ambiguous "longer forecast horizons are more difficult".

4.7 Explainability

A score on its own does not show what the model is responding to. This section uses SHAP values to attribute each prediction to individual features, and then checks that ranking against a second, independent importance measure.

SHAP values were computed on the test set to explain how much each feature contributes to the prediction. Currently a quench is by far the most important factor, then the adequation of the diet and GDP per capita. Comparing the two perceptions of time – the present and the past – one can observe a change: the weight of the current hunger factor has decreased from 0.272 to 0.217 while that of gdp per capita and population growth increased. Thus, the short-term risks depend mostly on the current situation; the long-term risks depend more on structural characteristics.

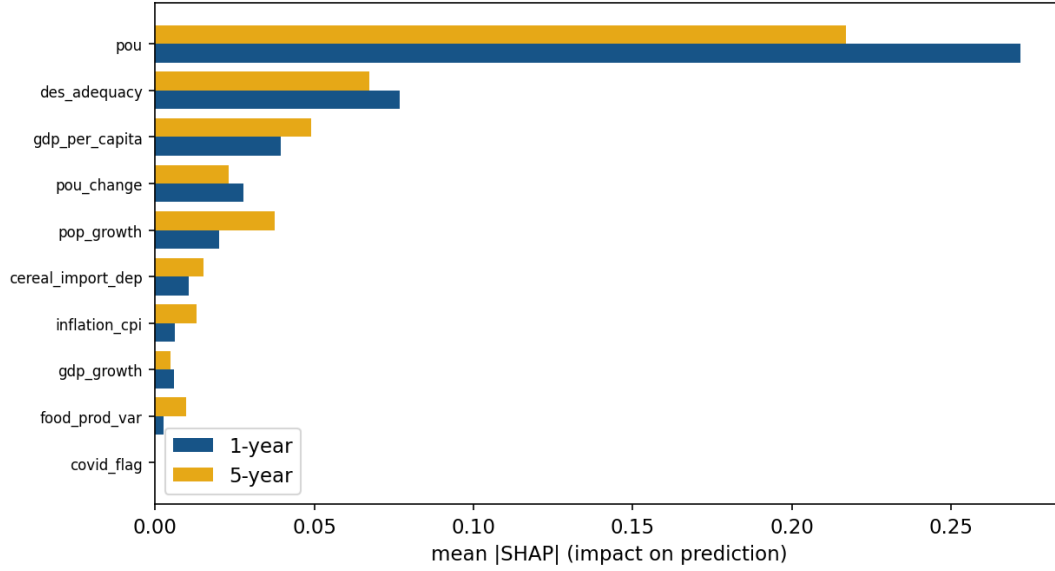


Figure 4. 2: SHAP feature importance for both horizons

Because Permutation importance was computed to serve as an independent measure of validity as ranking according to only one importance measure might be an implausible result. Table 4.7 shows the comparison of the two rankings. One current attribute, "current malnutrition," is ranked higher than any other features by both of the methods and two of the top three are consistent for both methods.

Feature	SHAP importance (1-year)	Permutation importance
pou	0.2717	0.5833
des_adequacy	0.0768	0.0039
gdp_per_capita	0.0394	0.0022
pou_change	0.0276	0.0045
gdp_growth	0.0060	0.0010

Table 4. 7: SHAP and permutation importance compared

4.8 Direction and Forecasting Results

The task of direction of change prediction is the hardest of the three because it lacks the clear information "most countries remain at the same risk level. The random forest in this case has a macro F1 score of 0.702 for the one-year forecast and 0.722 for the five-year forecast, against the

benchmarks of 0.276 and 0.202 respectively. This is why the increases the model obtains in this task are very high.

Horizon	Macro-F1	Accuracy	Baseline macro-F1
1 year	0.702	0.782	0.276
5 years	0.722	0.742	0.202

Table 4. 8: Risk-direction results

Five indicators were also forecasted using regression models for one and five years into the future. However, the one year ahead forecast of the malnutrition indicator was correct with an average absolute error of 0.54 percentage points. It was right at the persistence mark, as what was its mean? The largest R-squared was negative, by far, with the inflation number being the poorest forecast of all, and the negative R-squared is reported as a negative.

Indicator	Horizon	MAE	R-squared	Baseline MAE
pou	1yr	0.54	0.99	0.54
pou	5yr	2.12	0.88	2.12
gdp_per_capita	1yr	1556.54	0.98	1589.69
gdp_per_capita	5yr	2651.36	0.95	2710.57
inflation_cpi	1yr	5.8	0.54	5.85
inflation_cpi	5yr	9.86	-0.27	8.36
des_adequacy	1yr	1.04	0.99	1
des_adequacy	5yr	3.87	0.9	4.1
pop_growth	1yr	0.42	0.41	0.37
pop_growth	5yr	0.51	0.57	0.57

Table 4. 9: Forecast accuracy by indicator

4.9 Shock Propagation (RQ3)

RQ3 assesses the channels of transmission of a global shock via different variables. The data showed a clear macroeconomic shock in the case of the COVID-19 as the average GDP growth rate in the 2020 panel was -4.3% (see top panel of Figure 4.3). The question is whether the shock was reflected in the indicators of hunger and how much. Actually, at the level of the world, this was not shown. The average values of the malnutritional rates for all countries from 2010 to 2023 are plotted in the figure 4.3. The world-average reached its highest value of 10.22 in 2010, but hit a low in 2018 of 9.41. This data observed didn't reach a new high anymore after 2020. As the long-term trend was downward over the period, the shock's variation was under the average so averaging across 180 countries actually neutralized the effect.

When the definition is centered on risk categories (10) instead of averages, we would get a different picture. The composition of the risk categories for each year before and after the shock is presented in Table 4.10. Countries designated in 2019 as having a "high-risk" rose from 33 to 38 in 2021 and then decreased to 34 in 2023.

At the time of the shock, almost five countries would have been in the "high risk" category and the vast majority (but not all) of these countries would have been out of this category after the shock. This would have a smaller, but more targeted effect than the average effect per country indicates.

Year	Low	Medium	High
2019	80	52	33
2021	78	49	38
2023	80	48	34

Table 4. 10: Risk-tier composition before, during, and after the shock period

Figure 4.3 places the two series side by side: global GDP growth in the upper panel and global average hunger in the lower one, so the size of the economic shock and the muted response of the hunger series can be compared directly.

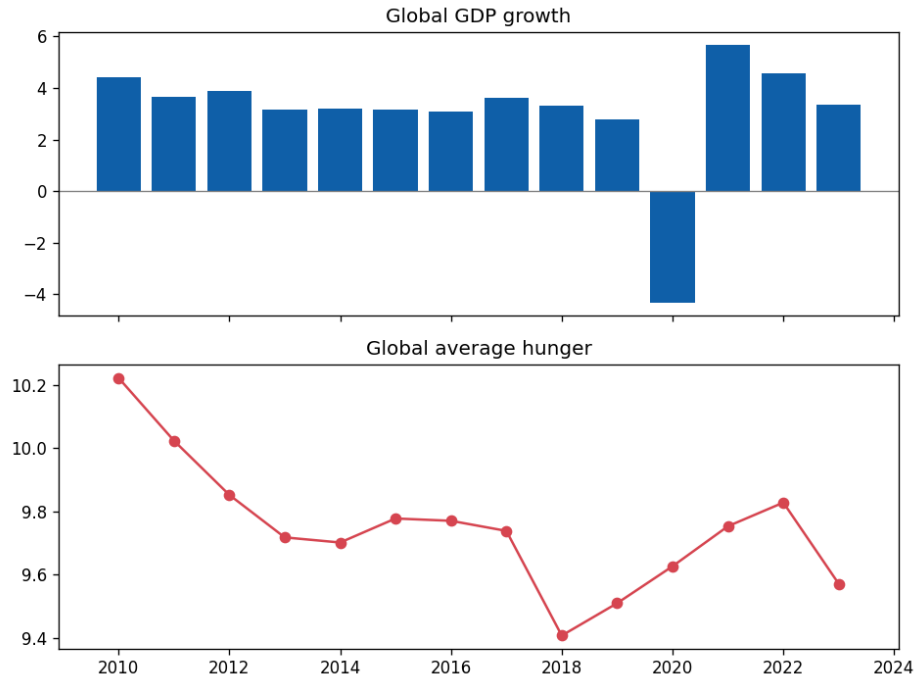


Figure 4. 3: Global GDP growth (upper) and global average hunger (lower), 2010-2023

The third piece of evidence is from the model itself. Because the model used indicators for 2020 and 2021 as part of its features so it could take advantage of them if it found them useful during the shock period, the model prior to fitting included these features; the SHAP importance for each time period is 0.0, indicating that the trained model did not use this indicator. Indeed, there was no indicator used in the fitted model but that of the present year. Again this evidence reinforces the idea that this indicator was only moved a small number of countries between the categories and not enough to change the linear relationship between this indicator and hunger scores in a way that would make it useful to the model.

The question of RQ3 is similar to RQ2 – what is the vulnerability of a country to hunger if it is more dependent on imports of grain? The median values of malnutrition as presented in Table 4.11 are given by countries when they are segmented on the basis of dependency on grain imports in quartiles. Does not rise monotonically: countries in the third quartile have higher hunger level than countries in the first quartile but there is a drop between the third and fourth (most import dependent) quartile; the maximum hunger country is not necessarily the most import dependent.

This deduction also applies in the case of RQ2: cereal import dependence is only positively correlated with high levels of hunger among countries that are also in low income levels, whereas the high dependent but prosperous ones can still achieve food security.

That is why the coefficient of cereal import dependence is only one of the strongest rules - a single correlation coefficient of 0.151.

Import-dependency quartile	Median pou (%)	Mean pou (%)	Countries
Q1 (lowest dependency)	2.50	5.52	487
Q2	7.10	11.42	486
Q3	7.65	11.11	486
Q4 (highest dependency)	5.80	8.95	487

Table 4. 11: Undernourishment by cereal import-dependency quartile

There is a limitation to the above analysis: for the analysis note, the lag between the shock and the response to hunger was assumed automatically by calculating the difference between the “year of the shock” and 2020. Since the global peak occurred in 2010 (at the start of the time series – and prior to the long-term declining trend in the data) this automatic calculation produced an invalid lag estimate and therefore no estimate is displayed in the notes. The responses of individual countries would have to be compared to determine the suitable lag for each country. This is recognised as a research area for the future in Section 5.3.

4.10 Findings

There are 4 tests, one for each research question. The dietary energy status measure is the one that has the highest correlation with hunger in these countries (corr = -0.83 ; OLS coefficient -0.5401 , $p < 0.001$), and it ranks second in the SHAP importance after current hunger, which alone accounts for 58.9% of total feature importance at one year and 49.7% at five years. In RQ2, optimal risk tier for next year (macro-F1 = 0.965) can be predicted across the 165 countries in the test set, but the results are not significantly different from choosing the country's current risk tier. The only

real benefit of the gold standard is that the explanation is given and tunable threshold (see above) is provided and there is no false positive error of High-to-Low. The COVID effect is not observed in the world average (which is strongly affected by the long term trends) for RQ3, but it is observed in the number of countries in the High tier in 2019-2021 and the number of countries in the Low tier in 2023, where flag COVID is not relevant for RQ3. Macro-F1 rate decreases for RQ4 from one year to five years, reflecting that there may not be enough training samples in the five-year horizon (not necessarily that the task of building a model is difficult); training set size curves indicate this is most likely an issue of not having enough training samples in the five-year horizon, and not hard cut-off. The most challenging – directions demonstrates the most stable growth from baseline.

4.11 Chapter Summary

This chapter examined how far the reported scores can be trusted. Beyond the headline macro-F1, the models were compared under a recall-weighted metric, inspected through the confusion matrix and per-class scores, and tested with threshold tuning, bootstrap confidence intervals and learning curves. Those checks qualify the results in two important ways: at the one-year horizon the advantage over the persistence baseline is not statistically significant, and the weaker five-year performance is explained by a shortage of training examples rather than by the model. SHAP and permutation importance explained what drives the predictions, and the shock analysis answered RQ3 by showing an effect that is invisible in the global average but visible in the count of countries entering the High tier.

CHAPTER (5)

CONCLUSION

This project demonstrated the application of the knowledge discovery (KDD) method to publicly available data from FAOSTAT and the World Bank in predicting food security risks at the national level for one- and five-year horizons. Descriptive mining resulted in dietary energy adequacy as the metric most strongly correlated with hunger and produced an association rule (confidence = 1.000) for "low income and insufficient supply" cluster. Predictive mining resulted in a random forest model with macro F1 scores of 0.965 (one-year horizon) and 0.865 (five-year horizon) and a directional classifier with F1 scores of 0.702 and 0.722 for each horizon, respectively, when using conditions significantly weaker than the baseline. Decisions were made exclusively through experimentation and the evaluation report discusses the performance potential of each.

5.1 Advantages of Using This Project

The points below summarise what the project delivers, each traceable to a result reported earlier in this book.

- These decisions are data-driven rather than hypothesis-driven: class balancing improved recall on the high flag from 0.953 to 0.974; ablation experiments indicated the effect of new features such as the `_filled` flag was negligible so were discarded.
- At both prediction times, there were no countries identified with high risk that were mistaken to have low risk-therefore, the most costly type of error in an early warning system.
- The threshold of the decision is also variable; this hypothesis is quantitatively validated – the recall of the "high" class obtained by the model on the test set when this threshold is 0.25 is 0.982.
- The prediction outcomes are explained by SHAP scores and cross-validated through permutation significance to avoid rankings to be based on one single technique.

- Outcomes are made available in an open workflow and dashboard, reproducible by a frozen image with an automated test suite.

5.2 Disadvantages of Using This Project and Limitations

The following limits are reported as measured, including the cases where the evidence does not support a strong claim.

- Since the confidence intervals do not overlap between the two methods when employing a one-year forecast horizon ([0.946, 0.979] and [0.944, 0.978]), the model does not outperform the benchmark model at this level of significance.
- For the five-year model, the problem is data paucity. With the largest training set, the learning curve has not leveled off and continues to grow as the model can learn from only 14 years of data from each country.
- Furthermore, inflation is mispredicted, with an R2 value having a negative value after five years which indicates that the model performs worse than a trivial model calculating the mean of this indicator.
- System works at the national level only; it does not explain the variation within the country, does not require the use of information on price, conflict or climate.
- Concurrent training of the random forest can cause small non-deterministic differences to be observed on the parallel machines and potentially influence the computation of important indicators in directional tasks.task.

5.3 Future Work

Each of the following follows directly from a limitation identified above rather than being a general suggestion.

- If indicators are skewed, apply a log transformation, which allows linear models to deal appropriately with these data, given that the normality tests indicate that no indicators are normally distributed.
- For each region, train models separately; regional differences in malnutrition rate are statistically significant according to the Kruskal-Wallis test.

- Extend over the history data current till 2010-2023, as the learning curve suggests the 5 year would gain most from the data.
- Use gradient boosting models for "Direction" task where the gradient boosting models showed the strongest improvements versus baseline models for this task.
- Calculate impact lag based on countries (individually) instead of globally, whereas the impact lag from averaging over 180 countries (Section 4.9) will hide the impacts of the hierarchical counts.

5.4 Chapter Summary

This chapter drew the work together. The project delivered a reproducible pipeline that predicts national food-security risk at two horizons, explains each prediction, and never labelled a High-risk country as Low on the test set. Its limits were reported as measured rather than minimised, including the statistically insignificant margin at one year and the data-starved five-year model. The future work listed follows directly from those limits, so the next steps are defined by the evidence rather than chosen arbitrarily.

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